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APPLY PROBLEM SOLVING STRATEGIES

US 14927

Section B: Summative assessment

Task 1

I. Thando's Print and copy shop in Kempton park has a problem where the business Local Area Network frequently disconnects. All 4 computers lose connection to the server, network printer and internet at random times during the day.

II. 1. Problem type: Network problem- Hardware and configuration
2. Problem parameters
•Occurs 4-5 times a day, mostly in the afternoon
•Affects all PCs, printer and internet a
•Each disconnection lasts 5-10 minutes
3. Possible causes
•Faulty or old Network switch
•Damaged Network cables
• IP address conflict
•Network overload from customers using business Wi-Fi

III. 1. Owner: "We lose customers when we can't print invoices. It happens most when we are busy. "
2. Staff: " We have to restart the router to get the internet back. It's frustrating.'
3.Observation: Cables are old and loose.
The switch is 6 years old. Customers are given Wi-Fi password.
Conclusion : Network instability is causing lost sales and productivity.

IV. Cross-cultural implications
1. Language : Staff speak English and Zulu. Training must use simple language and practical demos.
2. Tech literacy: Some staff do not know basic troubleshooting like checking cable connections.
3. Customer diversity: Customers expect fast service and Wi-Fi, which puts extra load on the network.

TASK 2

I. 1. Replace the old switch with a new 8-port Gigabit switch and replace damaged cables
2. Assign static IP addresses and create a separate Guest Wi-Fi network

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3. Upgrade internet package and router with ISP

II. 1. Cost- Affordable for a small business

2. Effectiveness - Solves the disconnect problem

3. Implementation time - How quickly it can be done

4. Maintainability - Easy for staff to manage after setup

III. Evaluate the 3 solutions*

Criteria	New Switch and cables	Static IP and Guest Wi-Fi	ISP Upgrade
Cost	Medium-R2500	Low-R0	High- R800 per month
Effectiveness	High	Medium	Medium
Implementation time	1 day	2 hours	1 week
Maintainability	Easy	Needs Training	Easy
Rating	Best Choice	Good Option	Too Expensive

Legislative considerations

1. POPIA: Separating Guest Wi-Fi protects customer and business data from unauthorized access.

2. OHS Act: New cables must be installed safely to avoid tripping hazards.